Python – Sample Study Notes

CampusAI demonstration material. These are sample notes; verify and expand them using your course syllabus.

What is Python?

- Python is a high-level programming language with readable syntax.
- It is used in web development, automation, data analysis, AI, and scripting.

Variables and Data Types

- A variable stores a value, for example: age = 20.
- Common types: int (whole number), float (decimal), str (text), bool (True/False).

Conditions

- if, elif, and else help a program make decisions.
- Example: if age >= 18: print('Adult')

Loops

- A for loop repeats over a sequence.
- A while loop repeats while a condition is true.
- Loops reduce repeated code.

Functions

- A function is a reusable block of code defined with def.
- Example: def greet(): print('Hello')
- Functions help organize and reuse code.